Tennessee Valley Authority, Post Office Box 2000, Spring City, Tennessee 37381

September 10, 2020
WBL-20-041

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Watts Bar Nuclear Plant, Units 1 and 2
Facility Operating License Nos. NPF-90 and NPF-96
NRC Docket Nos. 50-390 and 50-391

Subject: **Licensee Event Report 390/2020-003-00, Control Room Emergency Ventilation System Inoperable due to Main Control Room Door Being Left Open**

This submittal provides Licensee Event Report (LER) 390/2020-003-00. This LER provides details concerning the inoperability of the Control Room Ventilation System from a Main Control Room door being left open. This condition is being reported as an event or condition that could have prevented fulfillment of a safety function in accordance with 10 CFR 50.73(a)(2)(v)(D).

There are no regulatory commitments contained in this letter. Please direct any questions concerning this matter to Tony Brown, WBN Licensing Manager, at (423) 365-7720.

Respectfully,

Anthony L. Williams IV
Site Vice President
Watts Bar Nuclear Plant

Enclosure
cc: See Page 2

cc (Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Watts Bar Nuclear Plant

NRC FORM 366 (08-2020)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 08/31/2023

LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
Watts Bar Nuclear Plant, Unit 1	05000390	1 OF 5

4. Title
Control Room Emergency Ventilation System Inoperable due to Main Control Room Door Being Left Open

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
07	15	2020	2020	- 003 -	00	09	10	2020	Watts Bar Nuclear Plant, Unit 2	05000391
									NA	05000

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ **Other** (Specify here, in Abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include Area Code)
Charles Barker, Licensing Engineer	(423) 365-1771

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ **Yes** (If yes, complete 15. Expected Submission Date)		N/A	N/A	N/A

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

At 0835 Eastern Daylight Time (EDT) on July 15, 2020, a Main Control Room (MCR) alarm was received for low control room positive pressure. In response to the alarm, a Control Room Envelope (CRE) door was found ajar and immediately closed. Technical Specification Limiting Condition for Operation (LCO) 3.7.10, Control Room Emergency Ventilation System (CREVS), was declared not met for both trains and Condition B entered. At 0839 EDT on July 15, 2020, the alarm cleared, CREVS was declared operable and LCO 3.7.10, Condition B was exited.

This event was caused by a human performance error when an individual traversing the control building complex did not fully challenge and ensure a MCR envelope boundary door was properly latched and secured. The open control room door was identified and promptly closed. The worker was coached. Actions to prevent recurrence include briefing affected work groups, developing a video on door operation and to establish a dynamic learning activity for contract personnel that will be part of outage training.

This condition is being reported as an event or condition that could have prevented fulfillment of a safety function needed to mitigate the consequences of an accident in accordance with 10 CFR 50.73(a)(2)(v)(D).

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Watts Bar Nuclear Plant, Unit 1	05000-390	2020	- 003	- 00

NARRATIVE

I. Plant Operating Conditions Before the Event

Watts Bar Nuclear Plant (WBN) Unit 1 and Unit 2 were in Mode 1 at 100 percent rated thermal power (RTP).

II. Description of Event

A. Event Summary

At 0835 Eastern Daylight Time (EDT) on July 15, 2020, a Main Control Room (MCR) alarm was received for low control room positive pressure. In response to the alarm, a Control Room Envelope (CRE) door {EIIS:DR} was found ajar and immediately closed. Technical Specification (TS) Limiting Condition for Operation (LCO) 3.7.10, Control Room Emergency Ventilation System (CREVS) {EIIS:VI}, was declared not met for both trains and Condition B entered. At 0839 EDT on July 15, 2020, the alarm cleared, CREVS was declared operable and LCO 3.7.10, Condition B was exited.

This event is being reported to the Nuclear Regulatory Commission (NRC) under 10 CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented fulfillment of a safety function needed to mitigate the consequences of an accident.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

No inoperable structures, systems, or components contributed to this condition.

C. Dates and approximate times of occurrences

Date	Time (EDT)	Event
7/15/20	0835	Received MCR alarm 105-B and 106-B [MCR HVAC System A/B Abnormal]. In response to the alarm, Units 1 and 2 entered TS LCO 3.7.10, Control Room Emergency Ventilation System, CREVS, Condition B for one or more CREVS trains inoperable due to an inoperable CRE Boundary
7/15/20	0837	Discovered door C036 not fully closed. Door was able to be manually closed and latched.
7/15/20	0839	MCR Alarm 105-B and 106-B are clear. Units 1 and 2 exited TS LCO 3.7.10, CREVS, Condition B due to annunciators 105-B and 106-B clearing. CR 1622991 written to document the unplanned LCO entry.

D. Manufacturer and model number of each component that failed during the event

No equipment failures occurred during the event.

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

F. Method of discovery of each component or system failure or procedural error

Plant alarms indicated a loss of MCR positive pressure. The response procedure for low MCR pressure requires that the MCR doors be checked for proper closure, at which time door C036 was found open.

G. Failure mode, mechanism, and effect of each failed component

No equipment failures occurred during the event.

H. Operator actions

Upon receipt of the alarms, the MCR boundary door was promptly closed.

I. Automatically and manually initiated safety system responses

The MCR low pressure alarm properly actuated when the MCR door was left open.

III. Cause of the Event

A. Cause of each component or system failure or personnel error

No equipment failures occurred during the event.

B. Cause(s) and circumstances for each human performance related root cause

The event was caused by a human performance error when an individual traversing the control room boundary door failed to self check by fully challenging and ensuring the door was properly latched and secured.

IV. Analysis of the Event

The CRE is required to be operable in Modes 1 through 6 and during movement of irradiated fuel assemblies. Operability requires integrity of the CRE such that it will have a low unfiltered in-leakage during accident conditions to maintain the dose to operators within the requirements of Criterion 19 of 10 CFR 50, Appendix A. The TS’s allow the CRE boundary to be opened intermittently under administrative control, normally to allow routine personnel ingress and egress from the control room envelope. Administrative controls in the case of boundary doors are that an individual is in control of the door when it is opened.

On July 15, 2020, an individual traversing the control building complex did not fully challenge and ensure the MCR boundary door C036 was properly latched and secured. This action left this door ajar, resulting in operations personnel entering TS LCO 3.7.10, CREVS, for one or more CREVS trains inoperable due to an inoperable CRE boundary. Low positive pressure (less than 0.125 inches of water gauge WG) in the control room for 90 seconds results in a control room alarm. Upon receipt of the alarm, the CRE door was promptly closed. For this event, the CRE boundary was restored approximately four minutes after the MCR alarm was received. An engineering evaluation of a similar event is bounding for this event, and concludes that General Design Criteria (GDC) 19 dose limits to operators would not be exceeded when considering closure of the MCR door for accidents analyzed in the Updated Final Safety Analysis Report.

V. Assessment of Safety Consequences

A review of this event indicates, when considering the actual system capability and the response of equipment and personnel, a loss of safety function capable of impacting public health and safety did not occur with respect to the control room. This equipment is not analyzed in the site specific probabilistic risk assessment (PRA), but the impact of this door on an accident would be very small.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

The balance of the CRE equipment designed to protect the pressure boundary remained operable.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

Not applicable.

C. For failure that rendered a train of a safety system inoperable, an estimate of the elapsed time from the discovery of the failure until the train was returned to service

For this event, the MCR envelope door was closed within two minutes of receipt of the MCR alarm.

VI. Corrective Actions

This event was entered into the Tennessee Valley Authority's (TVA) Corrective Action Program and is being tracked under Condition Report (CR) 1622991.

A. Immediate Corrective Actions

The open control room door was identified and promptly closed. The worker was coached on this issue. A stand down was held with all contract personnel on proper door control which included video demonstrations. Also, the TVA contractor implemented a "two person" rule for entry into the MCR to ensure proper techniques are used, until September 30, 2020.

B. Corrective Actions to Prevent Recurrence or to reduce probability of similar events occurring in the future

Actions to prevent recurrence include briefing affected work groups, developing a video on door operation and to establish a dynamic learning activity for contract personnel that will be part of outage training.

VII. Previous Similar Events at the Same Site

LER 391/2020-001-00, reported an instance where the control room boundary door had been left open due to personnel error and promptly closed by operations in response to a low control room positive pressure alarm. The cause of this event is similar.

There have been other events in with the MRC doors such as LERs 390/2019-001-00, 390/2019-004-00, 390/2018-003-00, 390/2018-004-00, 390/2017-007-001 and 390/2017-014-00 with similar causes.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.